Adeel Razi

Turner Institute for Brain and Mental Health, School of Psychological Sciences
Monash Biomedical Imaging
Monash Data Futures Institute
Monash University, Melbourne, Australia
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EDUCATION

2012 PhD in Electrical Engineering
The University of New South Wales (UNSW), Australia.
Advisors: Jinhong Yuan (UNSW), Fellow IEEE, and Iain Collings (CSIRO), Fellow IEEE.

2006 M.Sc. in Communications Engineering
University of Technology Aachen (RWTH), Germany.
Advisors: Professor Gerd Ascheid (RWTH), Michael Speth and Herbert Dawid (Infineon Tech. AG).

2002 B.E. in Electrical Engineering
NED University of Engineering & Technology (NEDUET), Pakistan.
Rank: 1/140. Governor's Gold Medal.

CURRENT AND PAST APPOINTMENTS

2024 - to date Professor of Computational Neuroscience and AI

2020 - 2024 Associate Professor of Computational Neuroscience

2018 - 2020 Senior Lecturer and ARC DECRA Fellow
Turner Institute for Brain and Mental Health & Monash Biomedical Imaging
Monash University, Clayton, Australia.

2018 - to date Honorary Senior Research Fellow

2017 - 2018 Senior Research Fellow

2012 - 2017 Research Fellow
Wellcome Centre for Human Neuroimaging,
University College London, UK.
Supervisors: Professor Karl J. Friston, FRS & Professor Geraint Rees, FMedSci.

2011 - 2011 Staff Scientist
Broadcom Corporation, Sydney, Australia.

AWARDS and FELLOWSHIPS

5. Australian Research Council Future Fellowship, 2025-2029.
4. Fellow, Research Corporation for Advancement of Science, Scialog program on "Molecular Basis of Cognition" 2023-2026.
3. CIFAR Global Scholar 2021-2024. "Brain, Mind and Consciousness" (Directors: Anil Seth & Adrian Owen) & "Learning in Machines and Brains" (Directors: Yushua Bengio & Konrad Koerding).
2. Selected for the New York Academy of Sciences and Japan Agency for Medical Research and Development *Interstellar Initiative*.
1. Australian NHMRC Investigator Award as Emerging Leadership Fellow. 2021-2026.

RESEARCH FUNDING (Selected only as a Lead-PI), Total = $21 Million in past 5 years

7. Australian Research Council, Discovery Project, "Multi-modal generative models of brain function". Role: Lead-PI, 2026-2029. Value: $725,000. Co-I: Karl Friston.

6. Australian Research Council (ARC), Future Fellowship, "Multi-modal generative models of brain function". Role: Fellowship as Sole-PI, 2025-2029. Value: $1.285M.

5. Wellcome Leap program award on Untangling Addiction, "Psilocybin assisted therapy for opioid use disorder", Lead-PIs: Adeel Razi and Antonio Verdejo-Garcia, Co-Is: Suzanne Nielsen, Suresh Sundram. 2025-2027. $5.1M

4. Wellcome Leap program award on Untangling Addiction, "Neural network modelling informed personalized brain stimulation for alcohol use disorder", Lead-PIs: Antonio Verdejo-Garcia and Adeel Razi, Co-Is: Dan Lubman, Natalia Albein-Urios. 2024-2027. $9.2M

3. Office of National Intelligence (ONI), "Neural computers: Towards synthetic biological intelligence" Lead-PI: Adeel Razi, Co-Is: Chris French and Brett Kagan. 2023-2026. $599,760.

2. NHMRC Investigator Grant, "Understanding early stage neurodegeneration using computational modeling". Role: Fellowship as Sole-PI, 2021-2025. Value: $624,000.

1. Australian Research Council, Discovery Project, "Multiscale and multimodal modelling of brain dynamics". Role: Lead-PI, 2020-2023. Value: $525,000. Co-Is: Andrew Zalesky and Karl Friston.

PUBLICATIONS (Selected from 130+ Articles, Total Citations = 12,000+, H-index = 51)

10. D. Stoliker, L. Novelli, M. Khajehnejad, M. Biabani, T. Barta, M. D. Greaves,..., G. Deco, A. Seth, K. H. Preller, R. Carhart-Harris, S. Sundram, G. F. Egan, **A. Razi**, (2026) "Psychedelics Align Brain Activity with Context", *Nature*. To appear (accepted on 04 May, 2026). Link

9. M. Khajenejad+, F. Habibollahi, **A. Razi**, (2026) "BrainSymphony: A parameter-efficient multimodal foundation model for brain dynamics with limited data", *arXiv*. Link. World's first multimodal foundation model for brain imaging.

8. M.D. Greaves, L. Novelli, **A. Razi**, (2026) "Structurally informed resting-state effective connectivity recapitulates cortical hierarchy", *Proceedings of National Academy of Sciences*. To appear. Link

7. M.D. Greaves+, L. Novelli+, S. Mansour, A. Zalesky, **A. Razi**, (2025) "Structurally informed models of directed brain connectivity", *Nature Reviews Neuroscience*, vol.26, no. 1, pp. 23-41. Link

6. S. Ereira, S. Waters, **A. Razi**, C. Marshall, (2024) "Early detection of dementia with default-mode network effective connectivity", *Nature Mental Health*. Link

5. D. Stoliker+, L. Novelli+, F. Vollenweider, G. Egan, K. Preller, and **A. Razi**, (2024) "Neural mechanisms of resting-state networks and the amygdala underlying the cognitive and emotional effects of Psilocybin", *Biological Psychiatry*. Link

4. L. Novelli+, K. J. Friston, **A. Razi**, (2023) "Spectral Dynamic Causal Modelling: A Didactic Introduction and its Relationship with Functional Connectivity", *Network Neuroscience*. Link Highly cited paper as indexed by Clarivate.

3. B. J. Kagan, A. C. Kitchen, N. T. Tran, B. J. Parker, A. Bhat, B. Rollo, **A. Razi***, K. J. Friston*, (2021) "In vitro neurons learn and exhibit sentience when embodied in a simulated game-world", *Neuron*. Link. Media coverage in Nature, BBC, Economist, The Guardian, The Age, and other dozens of media outlets and blogposts. Ranked 1st based on altmetrics score across all papers ever published in Neuron and Ranked 1265 across all papers tracked by altmetrics.

2. L. Novelli+, **A. Razi**, (2022) "A mathematical perspective on edge-centric brain functional connectivity", *Nature Communications*. vol. 13, article no. 2693. Link

1. K. Rasheed+, A. Qayyum, J. Qadir, S. Sivathamboo, P. Kwan, L. Kuhlmann, T. J. O'Brien, **A. Razi**, (2021) "Machine learning for predicting epileptic seizures using EEG signals: A review", *IEEE Reviews in Biomedical Engineering*. vol. 14, pp. 139 - 155, 2021. Highly cited paper as indexed by Clarivate.

RESEARCH SUPERVISION

- **Postdoctoral Fellows**
  Current: 2 Senior Fellows (equivalent to Research Asst. Professor) and 7 Postdoctoral Fellows.
- **PhD Students**
  Current: 14 PhD students; 7 as the main supervisor and 7 as a secondary supervisor.
  Graduated: 6 PhD students, the first one graduating in 2020.
- **Masters and Bachelors (Honors) Students**
  Current: 1 Master's students; 3 Honors students. Graduated: 5 Master's and 22 Honor's students.

KEYNOTES and INVITED TALKS (Selected from 30+ in the past 5 years)

5. Keynote, AAAI 2026 workshop on NeuroAI, Singapore, Jan 2026.

4. Keynote at the Annual Meeting of the Organization for Human Brain Mappping, "Mechanisms of Mind: Bridging Generative Brain Models and Psychedelic Neuroscience", Brisbane, Australia, 2025.

3. Keynote, Caufield Grammar School, "Teaching brain cells to play computer games", Melbourne, Australia, Oct 2024.

2. Invited talk at the World Laureate Association as Young Scientist, "Towards Synthetic Biological Intelligence", Shanghai, China, Nov 2023.

1. Invited talk at the NSF AI Centre for Artificial and Natural Intelligence, Columbia University, "Active inference and DishBrain", June 2024. Host: Richard Zemel.

LEADERSHIP & KNOWLEDGE TRANSFER

1. Scientific Advisory Board
   - Cortical Labs, Australia
   - Active Inference Institute, USA
   - Sentient AI, USA
2. Chair of organizing committee of the First Workshop on NeuroAI at NeurIPS in Dec. 2024.
3. Founded an annual workshop on **"Maths in the Brain"** in 2019. It has since continued and this year was its 7th edition with 150+ attendees.
4. Chair, Organization for Human Brain Mapping, Australian Chapter. 2022-2025
5. Editorial Board Member for:
   - Imaging Neuroscience. 2026 - Present
   - PLoS Computational Biology. 2025 - present
   - Journal of Neuroscience (SfN). 2025 - present
   - Network Neuroscience (MIT Press). 2024 - Present
   - Scientific Reports (Nature). 2022 - Present
   - Computational Psychiatry. 2021 - Present
   - Frontiers in Neuroscience. 2020- present.
